

MEASURES OF SIMILARITY AND DISSIMILARITY

- Similarity and Dissimilarity are important because they are used by a number of data mining algorithms-clustering, nearest neighbour classification, clustering
- In many cases, initial data set is not needed once these similarities or dissimilarities have been computed
- Term 'Proximity' is used to refer to either similarity or dissimilarity

Similarity

- Numerical measure of the degree to which the two objects are alike
- Higher for pairs of objects that are more alike
- Usually non-negative and are often between 0 (no similarity) and 1 (complete similarity)

Dissimilarity

- Numerical measure of the degree to which the two objects are different
- Lower for more similar pairs of objects
- Sometimes fall in the interval $[0,1]$, but it is also common for them to range from 0 to ∞

Transformations

- Applied to convert a similarity to dissimilarity or vice versa
 - Example: We may have similarities ranging from 1 to 10, but an algorithm may work only with dissimilarities
- To transform a proximity measure to fall within a particular range
 - We may have similarities ranging from 1 to 10, but an algorithm may work only with similarities in the interval $[0,1]$

Example:

1.

- If the similarity falls in the interval $[0,1]$, then the dissimilarity can be defined as $d = 1 - s$

2.

- If similarities between objects range from 1(not at all similar) to 10 (completely similar), we can make them to fall within the range $[0,1]$ by using transformation $s' = (s - 1) / 9$
- In general, the transformation of similarities to the interval $[0,1]$ is given by $s' = (s - \min_s) / (\max_s - \min_s)$
- Similarly, the transformation of dissimilarities to the interval $[0,1]$ is given by $d' = (d - \min_d) / (\max_d - \min_d)$

Similarity and Dissimilarity between Simple Attributes

Table 2.7. Similarity and dissimilarity for simple attributes

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$
Ordinal	$d = x - y / (n - 1)$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - d$
Interval or Ratio	$d = x - y $	$s = -d, s = \frac{1}{1+d}, s = e^{-d},$ $s = 1 - \frac{d - \min_d}{\max_d - \min_d}$

Dissimilarity between data objects

Distances

- Dissimilarities with certain properties

Euclidean distance

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^n (x_k - y_k)^2},$$

- $\mathbf{x}$ and $\mathbf{y}$ are two points
- n is the number of dimensions and
- x_k and y_k are k^{th} attributes (components) of $\mathbf{x}$ and $\mathbf{y}$

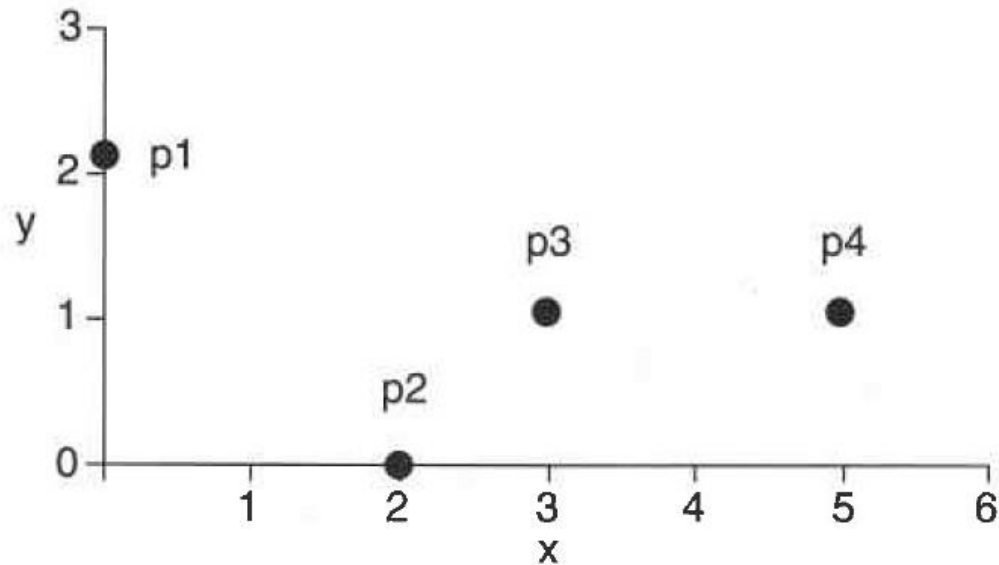


Figure 2.15. Four two-dimensional points.

Table 2.8. x and y coordinates of four points.

point	x coordinate	y coordinate
p1	0	2
p2	2	0
p3	3	1
p4	5	1

Table 2.9. Euclidean distance matrix for Table 2.8.

	p1	p2	p3	p4
p1	0.0	2.8	3.2	5.1
p2	2.8	0.0	1.4	3.2
p3	3.2	1.4	0.0	2.0
p4	5.1	3.2	2.0	0.0

The Euclidean distance measure is generalized by the Minkowski distance metric

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r},$$

where r is a parameter

Common examples of Minkowski distances

- $r = 1$. City block (Manhattan, taxicab, L_1 norm) distance. A common example is the **Hamming distance**, which is the number of bits that are different between two objects that have only binary attributes, i.e., between two binary vectors.
- $r = 2$. Euclidean distance (L_2 norm).
- $r = \infty$. Supremum (L_{max} or L_∞ norm) distance. This is the maximum difference between any attribute of the objects. More formally, the L_∞ distance is defined by Equation 2.3

$$d(\mathbf{x}, \mathbf{y}) = \lim_{r \rightarrow \infty} \left(\sum_{k=1}^n |x_k - y_k|^r \right)^{1/r}. \quad (2.3)$$

Table 2.10. L_1 distance matrix for Table 2.8.

L_1	p1	p2	p3	p4
p1	0.0	4.0	4.0	6.0
p2	4.0	0.0	2.0	4.0
p3	4.0	2.0	0.0	2.0
p4	6.0	4.0	2.0	0.0

Table 2.11. L_∞ distance matrix for Table 2.8.

L_∞	p1	p2	p3	p4
p1	0.0	2.0	3.0	5.0
p2	2.0	0.0	1.0	3.0
p3	3.0	1.0	0.0	2.0
p4	5.0	3.0	2.0	0.0

Properties of distances

If $d(x, y)$ is the distance between two points, x and y , then the following properties hold

1. Positivity

(a) $d(x, y) \geq 0$ for all x and y

(b) $d(x, y) = 0$ only if $x=y$

2. Symmetry

$d(x, y) = d(y, x)$ for all x and y

3. Triangle Inequality

$d(x, z) \leq d(x, y) + d(y, z)$ for all points x, y and z

- Measures that satisfy all three properties are known as **metrics**
- Many dissimilarities do not satisfy one or more of the metric properties
- Example: Non-metric Dissimilarities: Time

$$d(t_1, t_2) = \left\{ \begin{array}{ll} t_2 - t_1 & \text{if } t_1 \leq t_2 \\ 24 + (t_2 - t_1) & \text{if } t_1 \geq t_2 \end{array} \right\}$$

$d(1\text{PM}, 2\text{PM}) = 1 \text{ hour}$, while $d(2\text{PM}, 1\text{PM}) = 23 \text{ hours}$

Similarities between Data Objects

- If $s(x, y)$ is the similarity between points x and y , then the typical properties of similarities are:

1. $s(\mathbf{x}, \mathbf{y}) = 1$ only if $\mathbf{x} = \mathbf{y}$. ($0 \leq s \leq 1$)

2. $s(\mathbf{x}, \mathbf{y}) = s(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}$ and $\mathbf{y}$. (Symmetry)

Similarity Measures for Binary Data

Similarity coefficients

- Similarity measures between objects that contain only binary attributes
- typically have values between 0 and 1
- 1 indicates that the two objects are completely similar; 0 indicates that the objects are not at all similar
- Let x and y be two objects that consist of n binary attributes. The comparison of two such objects, i.e., two binary vectors, leads to the following four quantities (frequencies):

f_{00} = the number of attributes where x is 0 and y is 0

f_{01} = the number of attributes where x is 0 and y is 1

f_{10} = the number of attributes where x is 1 and y is 0

f_{11} = the number of attributes where x is 1 and y is 1

Simple Matching Coefficient

$$SMC = \frac{\text{number of matching attribute values}}{\text{number of attributes}} = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}}.$$

- Counts both presences and absences equally

Jaccard Coefficient

- Used to handle objects consisting of asymmetric binary attributes

$$J = \frac{\text{number of matching presences}}{\text{number of attributes not involved in 00 matches}} = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}.$$

$$\mathbf{x} = (1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$$

$$\mathbf{y} = (0, 0, 0, 0, 0, 0, 1, 0, 0, 1)$$

$f_{01} = 2$ the number of attributes where $\mathbf{x}$ was 0 and $\mathbf{y}$ was 1

$f_{10} = 1$ the number of attributes where $\mathbf{x}$ was 1 and $\mathbf{y}$ was 0

$f_{00} = 7$ the number of attributes where $\mathbf{x}$ was 0 and $\mathbf{y}$ was 0

$f_{11} = 0$ the number of attributes where $\mathbf{x}$ was 1 and $\mathbf{y}$ was 1

$$SMC = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}} = \frac{0 + 7}{2 + 1 + 0 + 7} = 0.7$$

$$J = \frac{f_{11}}{f_{01} + f_{10} + f_{11}} = \frac{0}{2 + 1 + 0} = 0$$

Cosine Similarity

- One of the most common measure of document similarity
- Can handle non binary vectors
- If $\mathbf{x}$ and $\mathbf{y}$ are two document vectors, then

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|},$$

- Where $\cdot$ Indicates the vector dot product
- $\|\mathbf{x}\|$ is the length of vector $\mathbf{x}$,

$$\mathbf{x} \cdot \mathbf{y} = \sum_{k=1}^n x_k y_k$$

$$\|\mathbf{x}\| = \sqrt{\sum_{k=1}^n x_k^2} = \sqrt{\mathbf{x} \cdot \mathbf{x}}$$

$$\mathbf{x} = (3, 2, 0, 5, 0, 0, 0, 2, 0, 0)$$

$$\mathbf{y} = (1, 0, 0, 0, 0, 0, 0, 1, 0, 2)$$

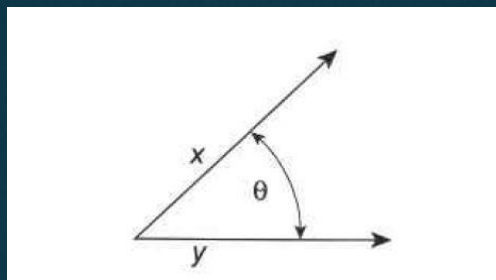
$$\mathbf{x} \cdot \mathbf{y} = 3 * 1 + 2 * 0 + 0 * 0 + 5 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 2 * 1 + 0 * 0 + 0 * 2 = 5$$

$$\|\mathbf{x}\| = \sqrt{3 * 3 + 2 * 2 + 0 * 0 + 5 * 5 + 0 * 0 + 0 * 0 + 0 * 0 + 2 * 2 + 0 * 0 + 0 * 0} = 6.48$$

$$\|\mathbf{y}\| = \sqrt{1 * 1 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 1 * 1 + 0 * 0 + 2 * 2} = 2.24$$

$$\cos(\mathbf{x}, \mathbf{y}) = \mathbf{0.31}$$

- cosine similarity really is a measure of the (cosine of the) angle between x and y



- if the cosine similarity is 1, the angle between x and y is 0 degree, and x and y are the same except for magnitude (length)
- If the cosine similarity is 0, then the angle between x and y is 90 degree, and they do not share any terms (words)

- Dividing $\mathbf{x}$ and $\mathbf{y}$ by their lengths normalizes them to have a length of 1

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}}{\|\mathbf{x}\|} \cdot \frac{\mathbf{y}}{\|\mathbf{y}\|} = \mathbf{x}' \cdot \mathbf{y}'$$

- For vectors with a length of 1, the cosine measure can be calculated by taking a simple dot product
- when many cosine similarities between objects are being computed, normalizing the objects to have unit length can reduce the time required

Difference between SMC and Jaccard index

The SMC is very similar to the more popular Jaccard index. The main difference is that the SMC has the term f_{00} in its numerator and denominator, whereas the Jaccard index does not. Thus, the SMC counts both mutual presences (when an attribute is present in both sets) and mutual absence (when an attribute is absent in both sets) as matches and compares it to the total number of attributes in the universe, whereas the Jaccard index only counts mutual presence as matches and compares it to the number of attributes that have been chosen by at least one of the two sets

In market basket analysis, for example, the basket of two consumers who we wish to compare might only contain a small fraction of all the available products in the store, so the SMC will usually return very high values of similarities even when the baskets bear very little resemblance, thus making the Jaccard index a more appropriate measure of similarity in that context. For example, consider a supermarket with 1000 products and two customers. The basket of the first customer contains salt and pepper and the basket of the second contains salt and sugar. In this scenario, the similarity between the two baskets as measured by the Jaccard index would be $1/3$, but the similarity becomes 0.998 using the SMC